MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.Tech. (CSE) MAKE UP SEMESTER EXAMINATION

SUBJECT: Data Communication (Programme Elective) – (CSE 4025)

Time: 3 hours **Date: 27-12-2019** **Max Marks: 50**

NOTE: DO NOT SEEK ANY CLARIFICATIONS FROM THE INVIGILATOR. MISSING DATA MAY BE ASSUMED SUITABLY.

Answer all the questions

1 a)	What is a data flow? Explain its various categories with suitable illustrations?	4 M
1 b)	What is the significance of mesh topology? Explain its advantages and disadvantages.	3 M
1 c)	Compare and contrast analog and digital signals with suitable diagrams?	3 M
2 a)	i) A non-periodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0 v. Draw the frequency domain of the signal. ii) Define signal to noise ratio with suitable schematic examples of large and small SNR.	4 M
2 b)	Define propagation and transmission time? What are the propagation time and the transmission time for a 5-Mbyte message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.	3 M
2 c)	What is baseline wandering? What are the implications of long runs of fixed amplitude?	3 M
3 a)	Compute NRZ, AMI, B8ZS and HDB3 line coding for the binary sequence 1010000110000000010.	4 M
3 b)	Draw the constellation diagrams for the following: i. ASK with a peak amplitude values of 1 and 3 ii. BPSK with a peak amplitude value of 2 iii. QPSK, with a peak amplitude value of 3 iv. 8-QAM with two different peak amplitude values, 1 and 3, and four different phases.	4 M
3 c)	The telephone line has 4 KHz bandwidth. What is the maximum number of bits we can send using each of the following techniques? Assume d = 0. **i.** ASK **ii.** QPSK	2 M

4 a) Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link with a neat diagram, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link. 5 M

4 b) Discuss the different kinds of propagation modes for unguided signals with a neat diagram. 3 M

4 c) Find the output port and the output VCI for packets with the following input port and input VCI addresses with respect to switch (router) in a virtual-circuit network shown in Fig.4c.1. 2 M
i) Packet 1: 3, 78 ii) Packet 2: 2, 92 iii) Packet 3: 4, 56 iv) Packet 4: 2, 71

Incoming		Outgoing	
Port	VCI	Port	VCI
1	14	3	22
2	71	4	41
2	92	1	45
3	58	2	43
3	78	2	70
4	56	3	11

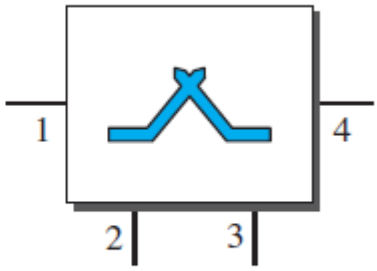


Fig 4c.1

5 a) Explain the different topologies that create a logical ring in a token passing network. 4 M

5 b) Explain in detail the transition phases of a point-to-point protocol with a neat diagram. 4 M

5 c) Enumerate the differences between unicast, multicast and broadcast addresses? 2 M